Group I Row III: the logarithm with a negative argument

$A \equiv (x-z)^2$, $B \equiv (x-w)^2$, $a \equiv \frac{k^+}{\vee}$. Blue = new, red = removed.

1 Your collaborator is right

In the term (Group I Row III, part 1, last line)

$$-\frac{ig^4 f^{abd}}{8\pi^5}\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2(x-w)^2(x-z)^2(y-z)^2}\log\Big[\frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2}\Big][\dots][\dots]$$

the argument $\frac{aB-A}{B}$ is negative whenever $(x-z)^2 > \frac{k^+}{\vee}(x-w)^2$. For $\vee \to \infty$ that is every configuration: the argument tends to $-\frac{(x-z)^2}{(x-w)^2} < 0$.

2 Where it comes from

The term comes from the third p^+ integral of your row (`Group_I.tex`, Row III, before "Integrating wrt p^+"):

$$\int_\Lambda^\vee dp^+\,\frac{k^+B+p^+A}{(p^+A-k^+B)\,p^+k^+}=\frac{1}{k^+}\int_\Lambda^\vee dp^+\Big[-\frac{1}{p^+}+\frac{2A}{p^+A-k^+B}\Big].$$

- The denominator $p^+A-k^+B=p^+(x-z)^2-k^+(x-w)^2$ vanishes at $p_0=\frac{k^+(x-w)^2}{(x-z)^2}$.
- p_0 lies inside $(\Lambda,\vee)$ exactly when $\Lambda A < k^+B < \vee A$. With $\Lambda\to 0$ this is $A > aB$, i.e. the region where your log is negative.
- There the integral passes through a simple pole. It has no value until a prescription is chosen.
- Writing the antiderivative as $\log(p^+A-k^+B)$ across the pole is what produced $\log\frac{k^+B-\vee A}{k^+B-\Lambda A}<0$.

Principal value.

$$\mathrm{PV}\int_\Lambda^\vee dp^+\,\frac{k^+B+p^+A}{(p^+A-k^+B)p^+k^+}=\frac{1}{k^+}\Big[-\log\frac{\vee}{\Lambda}+2\log\Big|\frac{\vee A-k^+B}{\Lambda A-k^+B}\Big|\Big]$$

Checked by numerical PV integration: equal to 10^{-14} (`pole_check.py`). With $\Lambda\to 0$ and the same split into $\log\frac{\vee}{k^+}$, $\log\frac{k^+}{\Lambda}$ as yours, the term becomes

$$\log\Big[\frac{\frac{k^+}{\vee}(x-w)^2-(x-z)^2}{(x-w)^2}\Big]\;\longrightarrow\;\log\Big|\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}\Big|\;\xrightarrow{\vee\to\infty}\;\log\frac{(x-z)^2}{(x-w)^2}.$$

The $\log\frac{\vee}{k^+}$ and $\log\frac{k^+}{\Lambda}$ terms you separated are unchanged.

3 The same in the other logs of Row III

Each log of Row III comes from a p^+ integral over one of the ranges below, with the same denominator. Its argument is negative exactly where p_0 lies in that range:

term (your `Other_terms.tex`)	from $\int dp^+$ over	negative where
part 1, line 2: $\log\frac{(x-z)^2-a(x-w)^2}{(x-z)^2-(x-w)^2}$	$(k^+,\vee)$	$aB<A<B$
part 1, line 3: $\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2}$	$(0,k^+)$	$A>B$
part 1, line 4 (this term): $\log\frac{a(x-w)^2-(x-z)^2}{(x-w)^2}$	$(0,\vee)$	$A>aB$
part 2, lines 3–8: $\log\frac{a(x-w)^2-(x-z)^2}{(x-w)^2-(x-z)^2}$	$(k^+,\vee)$	$aB<A<B$
part 3a: $\log\frac{(x-w)^2-(x-z)^2}{(x-w)^2}$	$(0,k^+)$	$A>B$

In all of them the principal value replaces the log by the log of the absolute value. My earlier checks of these terms (`UV_divergences.pdf`) used $\log|\cdot|$, i.e. the principal value.

4 The imaginary part does not cancel

If the denominator carries an $i\varepsilon$ (as a light-cone energy denominator does), each of these logs gets an extra $\mp i\pi\,\theta(p_0 \in \text{range})$ besides $\log|\cdot|$.

- The $i\pi$ terms cancel only if the sum of the residues at p_0 of all p^+ integrands of the row vanishes.
- With my transcription of the five integrands of your Row III, it does not vanish: at 8 random configurations the sums are between 10^{-2} and 0.6, with the separate residues of order 0.05–34 (`pole_check.json`).
- With the overall $i\,g^4 f^{abd}$, an $i\pi$ term becomes a real contribution $\propto \pi\,\theta\big((x-z)^2 > \frac{k^+}{\vee}(x-w)^2\big)$. It does not vanish for $\vee \to \infty$.

Which sign of $i\varepsilon$ (or the principal value) is correct must be read off the energy denominator of $\mathfrak{B}_3$ where Row III is derived; it cannot be fixed from `Other_terms.tex`.

Summary.

1. The negative argument is real: the p^+ integral runs through the zero of $p^+(x-z)^2 - k^+(x-w)^2$.
2. With a principal value, the fix is the absolute value: $\log\frac{(x-z)^2-\frac{k^+}{\vee}(x-w)^2}{(x-w)^2}$ in the region $(x-z)^2 > \frac{k^+}{\vee}(x-w)^2$, and the same for the other logs of the table.
3. With an $i\varepsilon$, there are in addition $\mp i\pi$ terms, which give a real, $\vee$-independent contribution to the row.